

# Cloud Computing, DevOps & AI

**01**

START HERE

Course syllabus and learning journey

**Lecture 1**

A practical foundation for building, shipping, and operating intelligent cloud systems.

# Why these three disciplines belong together

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Modern digital products need scalable platforms, reliable delivery, and intelligent capabilities.

01

## Cloud

Compute, storage, networking,  
and managed services.

02

## DevOps

Automation, collaboration,  
deployment, and observability.

03

## AI

Data-driven models and  
responsible intelligent features.

**The course connects these skills through a shared, hands-on project mindset.**

# By the end of the course, you will be able to...

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- Explain core cloud concepts and choose suitable service models.
- Use DevOps practices to build, test, release, and monitor software.
- Recognize where AI adds value—and the responsibilities that come with it.
- Connect technical choices to reliability, security, cost, and user impact.

**Our goal is practical fluency: enough depth to make sound decisions and build with confidence.**

# The syllabus moves from foundations to applied practice

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1

## Cloud foundations

Service models, deployment models, and core infrastructure.

2

## DevOps delivery

Version control, CI/CD, containers, and collaboration.

3

## Operating systems

Monitoring, reliability, security, and cost awareness.

4

## AI in the cloud

AI services, data, integration patterns, and ethics.

5

## Applied project

Design, build, and present a small cloud-enabled solution.

# How we will learn

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01

## Explore

Short lectures and  
guided discussion

02

## Practice

Labs, demos, and  
low-stakes exercises

03

## Apply

A project that  
combines the course  
concepts

04

## Reflect

Feedback,  
troubleshooting, and  
responsible design

**Participation matters: bring questions, try ideas, and learn from what does not work the first time.**

# Your first steps

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## Before our next session

- Review the course space and syllabus.
- Set up the tools named by your instructor.
- Write down one cloud, DevOps, or AI question you want answered.
- Come ready to connect the concepts to a problem you care about.

**Next: understanding the cloud landscape**